

Asymmetric Allylic Substitutions Using Organometallic Reagents

Valentín Hornillos, Jean-Baptiste Gualtierotti, and Ben L. Feringa

Abstract This chapter summarises recent progress in Cu-catalysed asymmetric allylic alkylation (AAA) with organometallic compounds, including Grignard, organolithium, organoaluminium, organozinc and organozirconium reagents. New reaction conditions and chiral ligands that improve these transformations or allow to overcome previous limitations associated with chemo-, regio- and enantioselectivities will be described. Moreover, a description of new ligands and conditions for the introduction of previously elusive nucleophiles, such as highly reactive organolithium compounds, is included together with a brief mechanistic discussion. Additionally, new challenging substrates which provide densely functionalised building blocks, as well as new synthetic applications that take advantage of the terminal olefin formed in these reactions, will be described.

Keywords Allylation • Asymmetric allylic substitution • Copper catalysis • Enantioselectivity • Organometallic reagents

Contents

- 1 Introduction
 - 2 Asymmetric Allylic Substitutions Using Grignard Reagents
 - 3 Asymmetric Allylic Substitutions Using Organolithium Reagents
 - 4 Asymmetric Allylic Substitutions Using Organoaluminium, Zinc and Zirconium Reagents
 - 5 Asymmetric Allylic Substitutions on Racemic Substrates with Organometallic Reagents
- References

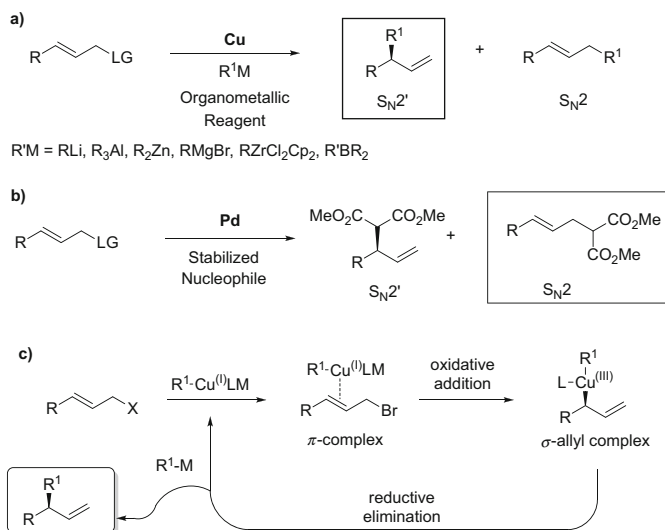
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1 Introduction

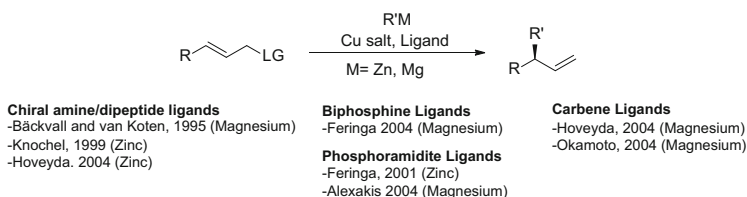
Asymmetric catalysis using transition metals is a highly efficient tool for the construction of stereogenic centres with control over the absolute configuration, which is often a key factor determining the activity of pharmaceuticals [1–3]. This strategy presents various advantages such as atom economy, reduction of by-product formation, functional group tolerance and the capability to provide the desired product with high levels of selectivity. Several transition metals have proven their efficiency in catalytic asymmetric processes [4]. In particular, the use of copper provides cheaper and more readily available catalyst systems while maintaining high activity and selectivity. Among the Cu-catalysed enantioselective processes, asymmetric allylic alkylation (AAA) represents a useful and powerful methodology for the synthesis of chiral building blocks, in particular for the synthesis of pharmaceuticals and natural products [4–12]. This transformation generally leads to the S_N2' -functionalised product, through the reaction of a nucleophile and an allylic system, bearing a suitable leaving group, and provides a carbon-based stereocentre next to a terminal olefin (Scheme 1a). This alkene moiety can be further converted into a variety of functionalities while preserving the stereochemical integrity of the starting material. Cu-catalysed AAA represents a highly efficient catalytic process for the use of organometallic reagents, such as organozinc, organomagnesium, organoaluminium, organozirconium and organolithium compounds. These AAA reactions, with these “hard” C-nucleophiles, are complementary to the well-known Pd-catalysed asymmetric allylic alkylations which employs soft and stabilised nucleophiles [13–16]. These nucleophiles generally approach the Pd-allyl moiety from the less hindered side of the allyl moiety coordinated to the metal ion leading to the S_N2 product (Scheme 1b). In the case of Cu-catalysed AAA, the organometallic compound first undergoes transmetalation to a Cu (I) complex, followed by π -complex formation and subsequent oxidative addition to form a Cu (III) σ -allyl complex. This is also the enantiodiscriminating step. Finally, reductive elimination gives rise to the new C–C bond and the branched (S_N2') product (Scheme 1c) [17].

The groups of Bäckvall and Van Koten pioneered the use of hard nucleophiles in their Cu-catalysed AAA of Grignard reagents and allylic acetates [18–21]. Knochel's work [22] illustrated that bulky dialkylzinc reagents were also suitable nucleophiles, and the groups of Feringa [23, 24] and Alexakis [25] subsequently reported on the use of phosphoramidite ligands in combination with linear dialkylzinc and Grignard nucleophiles featuring excellent (up to 99% ee) selectivities. Later on, a variety of peptide-based ligands, aminoalkyl phosphites or N-heterocyclic carbene (NHC) ligands, in combination with organozinc, Grignard and organoaluminium compounds, have also been shown to be effective in the Cu-catalysed AAA of a variety of di- and trisubstituted allyl systems (Scheme 2). As several reviews have already been reported on this field [6–12], this chapter will focus on developments in Cu-catalysed AAA since 2010.

Asymmetric Allylic Substitutions Using Organometallic Reagents



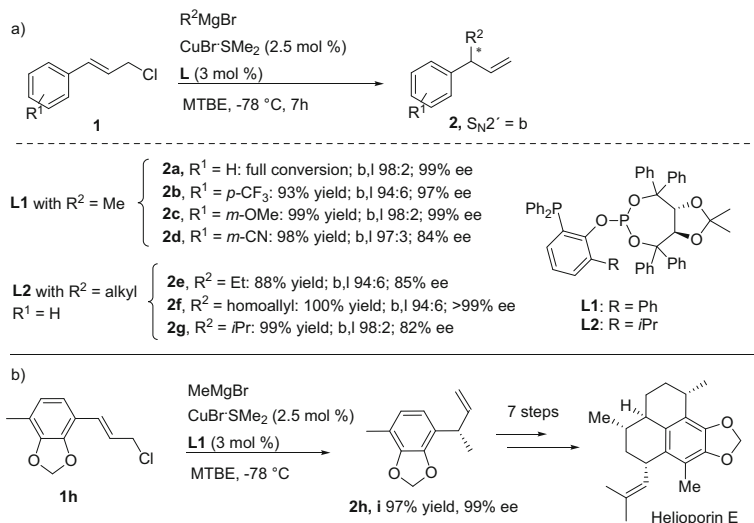
Scheme 1 (a) Pd- and (b) Cu-catalysed AAA methodologies. (c) General mechanism on the Cu-catalysed AAA



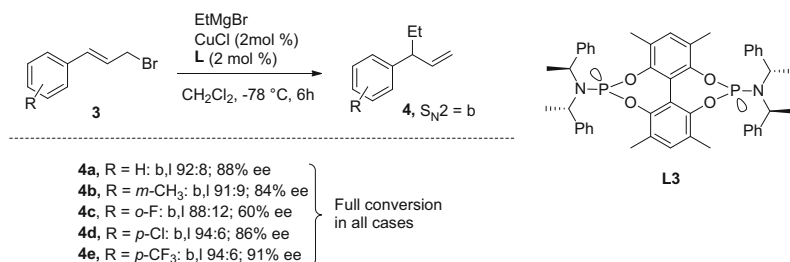
Scheme 2 Early reports on Cu-catalysed AAA methodologies

2 Asymmetric Allylic Substitutions Using Grignard Reagents

Schmalz and co-workers screened a library of modular phosphine–phosphoramidites as chiral ligands for the copper-catalysed allylic alkylation of cinnamyl chlorides **1** and Grignard reagents resulting in a broader methodology than previously reported [26]. In general, better selectivities were found for TADDOL-derived ligands in comparison with BINOL-derived ligands, and **L1** in combination with $CuBr \cdot SMe_2$ was found optimal for the addition of $MeMgBr$ (Scheme 3a). The use of less coordinating ethereal solvents, in particular MTBE, yielded the highest regio- and enantioselectivities (S_N2'/S_N2 ratio up to 98:2; ee up to > 99%). Modest enantioselectivities were obtained when using longer alkyl Grignard reagents although by switching to related phosphine–phosphite ligand **L2** the enantioselectivity was partially restored. The method was further employed



Scheme 3 Copper-catalysed AAA of cinnamyl chlorides with Grignard reagents employing, as chiral ligands, phosphine–phosphoramidites

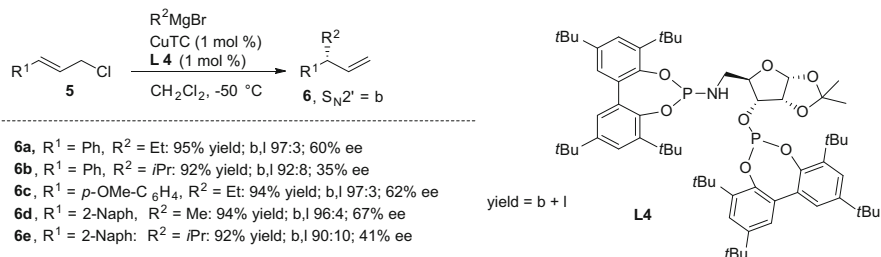


Scheme 4 Copper-catalysed AAA of cinnamyl chlorides with EtMgBr employing an atropis bridged phosphoramidite

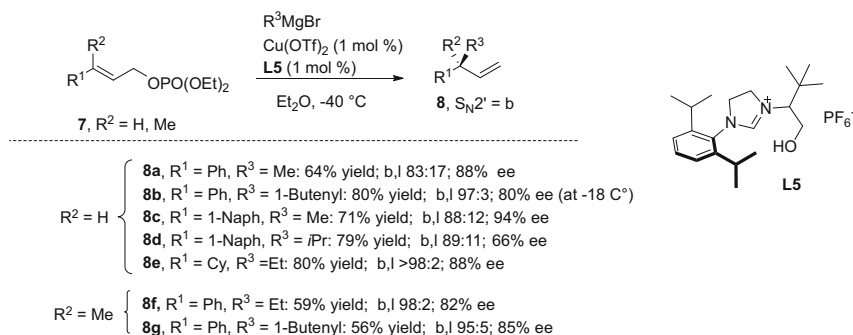
to set up the first stereogenic centre in the enantioselective total synthesis of *helioporins C* and *E*, which are marine metabolites with antiviral and cytotoxic activity (Scheme 3b) [27].

At the same time, the group of Zhang independently showed that D_2 symmetric atropis biphenyl phosphoramidites were efficient ligands for the Cu-catalysed AAA of cinnamyl bromide derivatives **3** [28]. These ligands have two identical substituents linked pairwise by four bridges, which induces the axial chirality. Tetramethyl-substituted **L3** promoted the addition of EtMgBr with good regioselectivity and moderate to good enantioselectivity (Scheme 4). However, reactions with other Grignard reagents were not reported.

The group of Alexakis subsequently reported a study on the use of bidentate furanoside phosphite–phosphoramidite and diphosphoramidite ligands for the Cu-catalysed AAA of cinnamyl-type substrates [29]. A range of different cinnamyl



Scheme 5 Cu-catalysed asymmetric allylic alkylation using a furanoside phosphite-phosphoramidite as chiral ligand **L4**

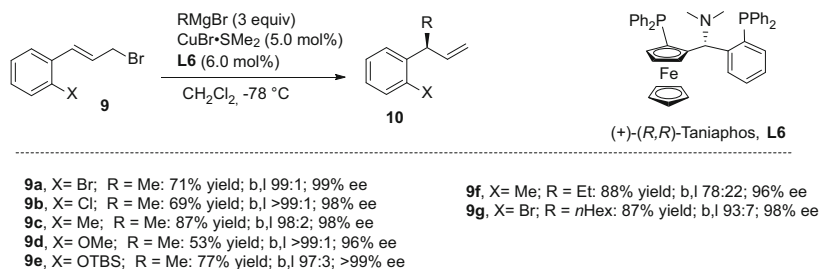


Scheme 6 Bidentate hydroxyalkyl NHC ligand for the copper-catalysed AAA

chlorides **5** and Grignard reagents could be coupled, with only 1 mol% of **L4** and CuTC as the catalyst system, with excellent regioselectivity, but modest enantioselectivity (Scheme 5).

Mauduit and co-workers investigated the use of bidentate hydroxyalkyl NHC ligands, derived from the amino acid chiral pool, in the Cu-catalysed AAA of cinnamyl phosphates **7** with alkyl Grignard reagents [30]. Using **L5** as ligand precursor, a variety of linear Grignard reagents afforded good regio- ($\text{S}_{\text{N}}2'/\text{S}_{\text{N}}2$ up to > 98:2) and enantioselectivities in the alkylation reaction, especially using MeMgBr (up to 94% ee). A practical advantage is that the Grignard reagent also deprotonates the imidazolium salt **L5**, without the necessity of using an external base or preformed copper or silver carbene complex. Diminished enantioselectivity was found in the case of secondary Grignard reagents (Scheme 6). Moreover, the reaction conditions could be extended to the formation of all-carbon quaternary stereocentres in high yields and with good levels of enantioselectivity.

The use of *ortho*-substituted cinnamyl substrates in Cu-catalysed AAA is a particularly difficult transformation, especially when the reaction is performed with MeMgBr, as an erosion on the regio- and enantioselectivity is generally observed. Feringa and co-workers demonstrated that, by employing their previously developed copper-taniaphos (**L6**) catalyst, a variety of *ortho*-substituted cinnamyl



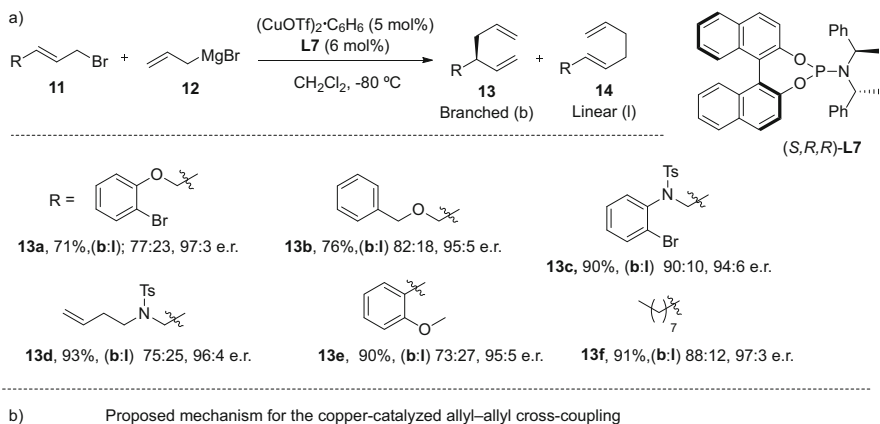
Scheme 7 Copper-catalysed AAA of *ortho*-substituted cinnamyl bromides **9** with Grignard reagents

substrates **9** can participate in the alkylation reaction with alkyl Grignard reagents, affording the corresponding branched products **10** with excellent regio- and enantioselectivity (S_N2'/S_N2 up to >99:1 and up to 99% ee) [31]. A variety of *ortho* substituents including Br, Cl, CH₃, -OMe and OTBS were tolerated in the reaction with MeMgBr as well as longer alkyl Grignard reagents (EtMgBr and *n*-HexMgBr) when using *ortho* methyl- and *ortho* methoxy-cinnamyl bromide as substrates (Scheme 7).

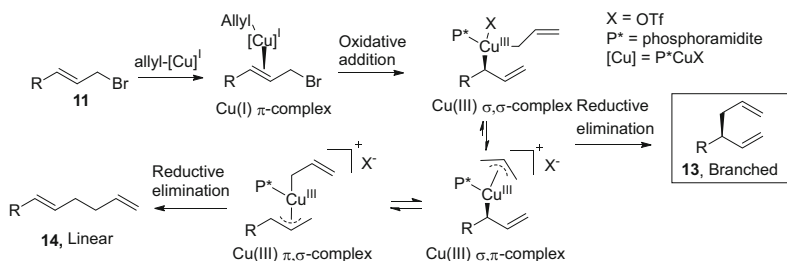
Feringa and co-workers used (*R,S,S*)-**L7** as ligand to promote the first Cu-catalysed asymmetric allylic alkylation using an allyl Grignard reagent [32]. Although this reaction is similar to the conventional Cu-catalysed AAA with alkyl nucleophiles, the dual coordination modes of allyl metal species may have delayed the development of an effective enantioselective version of this reaction. The use of allylMgBr in the reaction with allyl bromides **11** afforded chiral 1,5-dienes **13** in good yields and high enantioselectivity (Scheme 8a). The use of non-coordinating counter anions in the copper salt precursor favours the branched substitution product **13**, with (CuOTf)₂·C₆H₆ being optimal. Excellent enantioselectivities were obtained for a variety of substrates bearing different functionalities including protected alcohols, amines, alkenes or acetals. The authors proposed that the corresponding Cu^{III} σ–σ complex, which is formed after oxidative addition of an allyl-Cu**L7**OTf to an allyl bromide, is in equilibrium with a Cu^{III} σ–π intermediate. Fast reductive elimination from these intermediates would produce the branched substitution product. However, due to stabilisation caused by the η^3 bonding mode of the allyl group, isomerisation from the Cu^{III} σ–π-complex to the Cu^{III} π–σ-complex may occur, giving rise to the formation of the linear product. This proposal is further supported by the fact that for cinnamyl-type substrates, lower regioselectivity is observed. In this case, the presence of an aryl ring in the benzylic position may further stabilise the Cu^{III} σ–π-complex, favouring the above-mentioned isomerisation (Scheme 8b).

In 2011 Tomioka and co-workers expanded their previously reported Cu-catalysed asymmetric allylic arylation of aryl Grignard reagent and cinnamyl bromides [33] to the corresponding aliphatic bromides [34]. For this purpose, structural steric and electronic modifications of their chiral Cu(I)-NHC **C1** catalyst

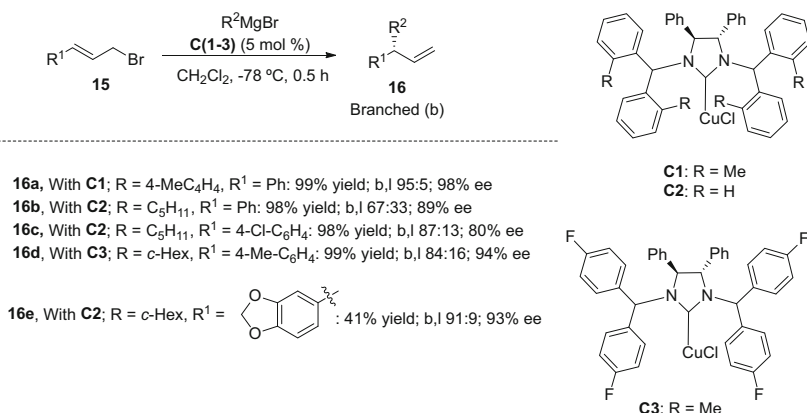
Asymmetric Allylic Substitutions Using Organometallic Reagents



b) Proposed mechanism for the copper-catalyzed allyl-allyl cross-coupling

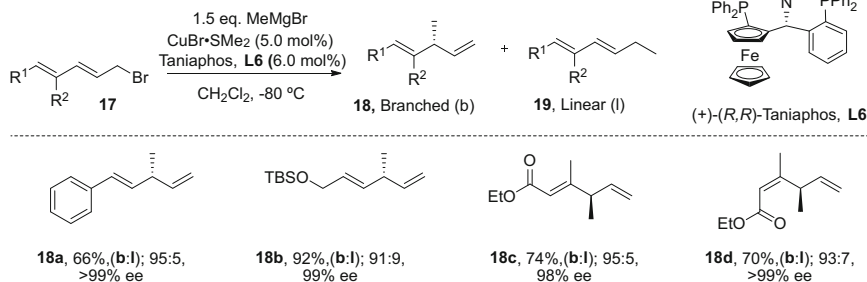


Scheme 8 Asymmetric allylic alkylation using allylMgBr

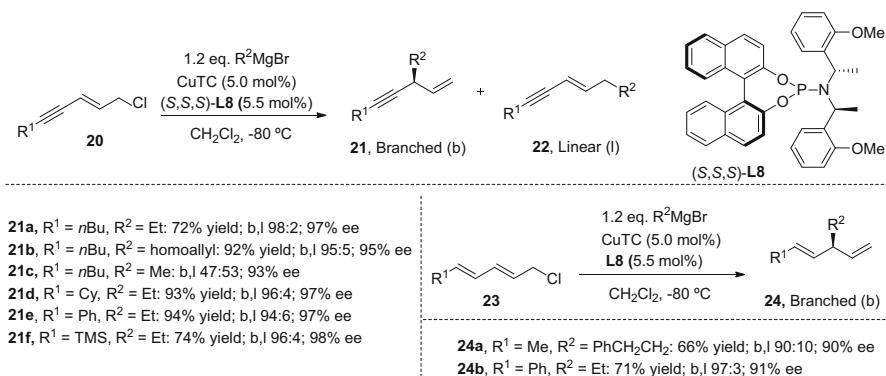


Scheme 9 Asymmetric allylic arylation of aliphatic allylic bromides

were carried out. They found that the presence of benzhydryl (**C2**) or 4,4'-difluorobenzhydryl (**C3**) groups at the N atoms of the catalyst allowed to obtain moderate to high S_N2' selectivity ($S_{N2'}/S_{N2}$ ratios up to 96:4) with good to excellent



Scheme 10 Synthesis of skipped dienes **18** via Cu-catalysed AAA with MeMgBr

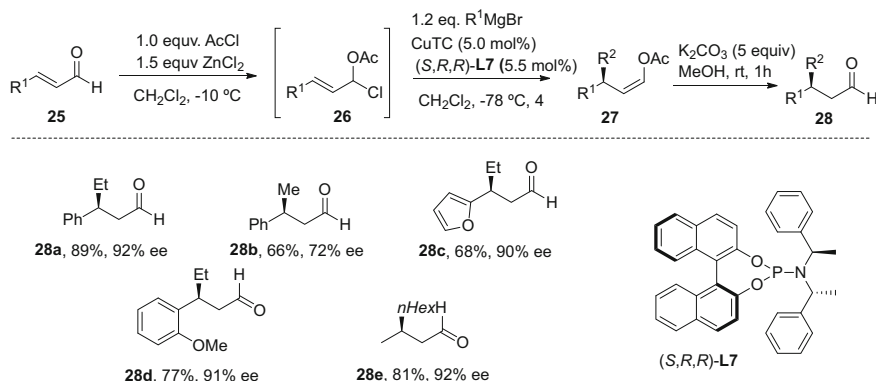


Scheme 11 Synthesis of chiral 1,4-enynes and 1,4-dienes via Cu-catalysed AAA with alkyl Grignard reagents

enantioselectivities (up to 96% ee) in the reaction of a variety of bulky or electron-poor aryl Grignard reagents and aliphatic allylic bromides **15** (Scheme 9).

Recently, Feringa and co-workers have demonstrated that chiral 1,4-dienes **18** (skipped dienes) can be prepared in excellent regio- and enantioselectivities (S_N2'/S_N2 ratio up to 97:3 and up to 96% ee) via the Cu-catalysed asymmetric allylic alkylation of prochiral diene bromides **17** with MeMgBr using Taniaphos **L6** as chiral ligand [35]. This is an important transformation as many biologically active molecules as *hennoxazole A*, *ansalactam A*, *lejimalide* and *phorbosins* present this motive in their structures. Notably, the *E*-geometry of the double bond was found crucial to obtain high regio- and enantioselectivity. However, these values are independent of the geometry of the remote double bond as similar selectivities were obtained by using a (*Z,E*)- or (*E,E*)-diene **17** (Scheme 10).

Using phosphoramidite ligand **L8** allowed Alexakis and co-workers to perform a similar transformation but employing prochiral *E*-enyne chlorides **20** as substrates [36]. A variety of alkylmagnesium bromide reagents were used as nucleophiles providing 1,4-enynes **21** in excellent regio- and enantioselectivities (S_N2'/S_N2 ratio up to 98:2 and up to >99% ee). Unlike in the Feringa report, the use of MeMgBr

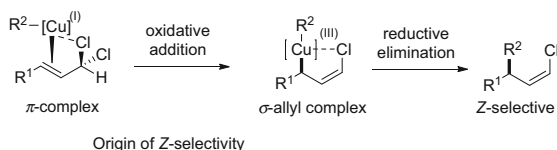
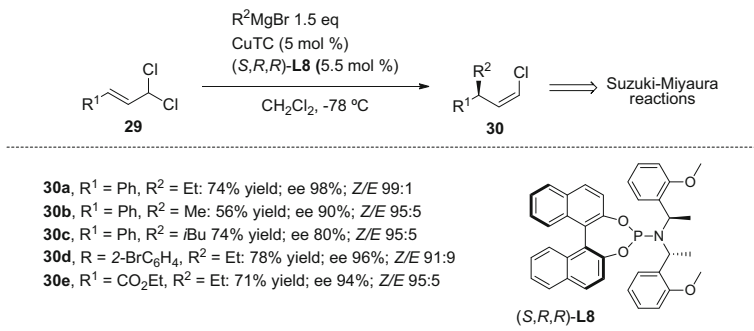


Scheme 12 Cu-catalysed enantioselective synthesis of chiral enol acetates **27** and β -substituted aldehydes **28**

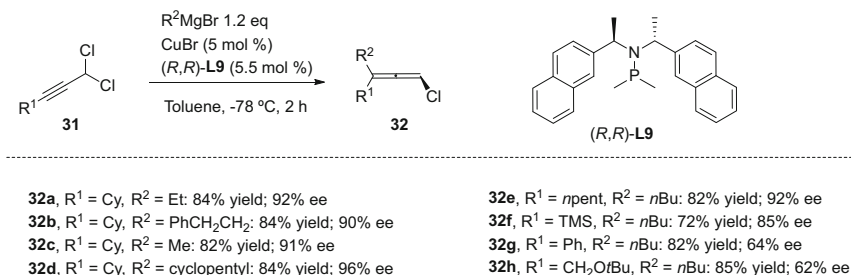
constitutes a limitation in terms of regioselectivity (S_N2'/S_N2 ratio 1:1), although the enantioselectivity was maintained (Scheme 11). This method can be considered complementary to the Cu-catalysed AAA using alkynylaluminium reagents, recently described by Hoveyda and co-workers [37], as both lead to 1,4-enynes, although with reversed polarity of the reaction partners. Similar to Feringa's observations, the selectivity of the reaction substantially decreases when the *Z* isomer was used. Additionally, the scope of the reaction was also extended to conjugated *E,E*-diene chlorides **23** which showed to behave similarly, affording the chiral 1,4-dienes **24** in high yield and selectivity (Scheme 11).

Asymmetric allylic alkylation reactions of geminal disubstituted allylic electrophiles have been studied by the group of Feringa. In 2010, they reported that the use of the phosphoramidite ligand (*S,R,R*)-**L7**, in the Cu-catalysed reaction between Grignard reagents and α -chloroallylic acetates **26**, in situ prepared from the corresponding α,β -unsaturated aldehydes **25**, affords enantiomerically enriched enol acetates **27** in a one-pot process (Scheme 12) [38]. Exposure of these products to methanolysis under basic conditions delivered the products of the formal conjugate addition to β -substituted aldehydes **28**. This reaction represents a viable alternative to the conjugate addition to enals, which is problematic due to the reactivity of the carbonyl group towards these organometallic compounds. The transformation affords excellent regio- and enantioselectivities (up to 94% ee) with a variety of primary alkyl Grignard reagents. The conventional limitation of phosphoramidites for the Cu-catalysed AAA of methyl groups is again manifested, as a decrease of regio- and enantioselectivity was observed for the use of MeMgBr.

The high *Z*-selectivity observed in the formation of the enol acetate encouraged Feringa and co-workers to investigate the substitution of allylic *gem*-dichlorides **29** (Scheme 13) [39]. The use of (*S,S,S*)-**L8** in the alkylation reaction with a variety of primary Grignard reagents affords chiral *Z*-vinyl chlorides **30** with excellent regio-, enantio- and *Z:E*-selectivity (99:1 *Z:E* ratio, ee up to 98%). The use of an ester-



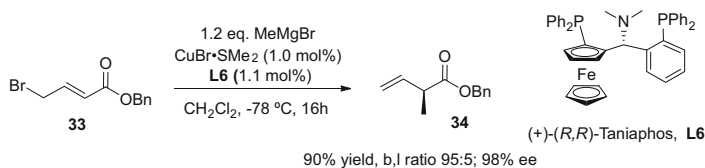
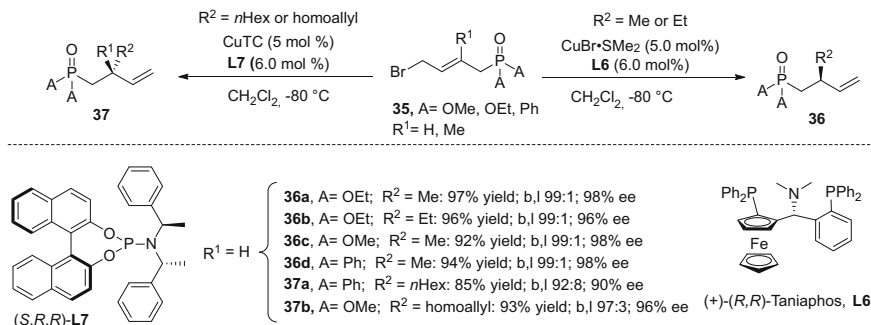
Scheme 13 Z-Selective Cu-catalysed asymmetric allylic alkylation



Scheme 14 Copper-catalysed asymmetric propargylic alkylation using Grignard reagents

substituted dichloride is also well tolerated, without the formation of carbonyl addition or S_N2 products. The corresponding Z-vinyl chlorides **30** were subsequently employed in Suzuki–Miyaura cross-coupling reactions providing a variety of chiral Z-alkyl- and aryl-substituted alkenes and dienes. To explain the selectivity for the (Z)-isomer, the authors propose that an extra coordination between the copper complex and one of the chlorides favours the formation of a (Z)-σ allyl complex that subsequently undergoes (Z)-selective reductive elimination to give the alkylated product (Scheme 13).

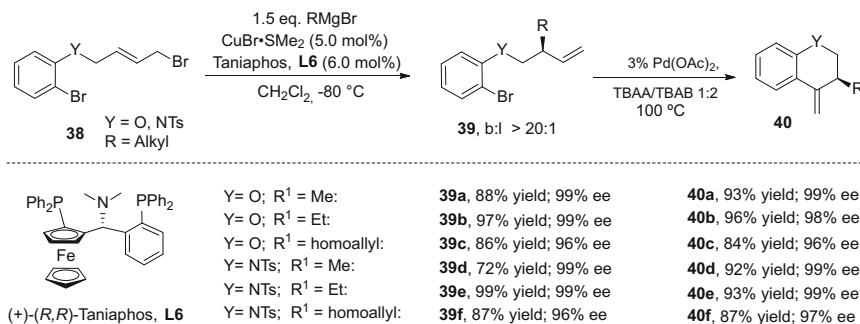
A similar system, but employing prochiral propargylic compounds, has been reported by Alexakis and co-workers [40]. In this case, 1,1-dichloro propargylic compounds **31** prepared from the corresponding aldehydes were reacted with a variety of primary and secondary Grignard reagents using copper-SimplePhos complex as catalyst. A variety of chiral allenenes **32** were obtained with high regioselectivities and good enantioselectivities (up to 96% ee) (Scheme 14). Unlike in the copper-catalysed AAA using MeMgBr and other phosphoramidites, the

**Scheme 15** Cu-catalysed AAA of halocrotonates**Scheme 16** Cu-catalysed AAA phosphonate and phosphine oxide allylic bromides

methyl group could be successfully introduced with good regio- and enantioselectivity. Transformation of the final products into the corresponding alkylated allenes or arylated propargylic compounds was also performed showing complete chirality transfer.

A variety of Cu-catalysed AAA reactions with allylic electrophiles, bearing synthetically useful functionalities, have been described by the group of Feringa. They first reported the enantioselective synthesis of α -methyl-substituted ester **34** via allylic alkylation of benzyl 4-bromocrotonate **33** using copper–Taniaphos **L6** as chiral ligand (Scheme 15) [41]. The corresponding products were further elaborated into chiral multifunctional building blocks including hydroxyl- or iodo-substituted lactones, without significant loss of stereochemistry.

Using the same ligand **L6**, this group also described the enantioselective copper-catalysed alkylation of phosphonate and phosphine oxide allylic bromides **35** with methyl- and ethylmagnesium bromide in good yields and high regio- and enantioselectivity (S_N2'/S_N2 up to 99:1 and up to 98% ee) (Scheme 16) [42]. However, with this ligand, the use of Grignard reagents with a longer alkyl chain afforded more moderate values. Further screening showed that phosphoramidite (S,R,R)-**L7**, in combination with *n*-hexyl- or homoallyl magnesium bromide, restored the selectivity towards the desired products **37** (S_N2'/S_N2 up to 97:3 and up to 96% ee). This reaction constitutes an alternative to the catalytic enantioselective conjugate addition of alkyl nucleophiles to α - β -unsaturated organophosphorus compounds, which has not been described to date. The reaction was further extended to the enantioselective synthesis of all-carbon quaternary centres although the



Scheme 17 Tandem asymmetric allylic alkylation and intramolecular Heck coupling

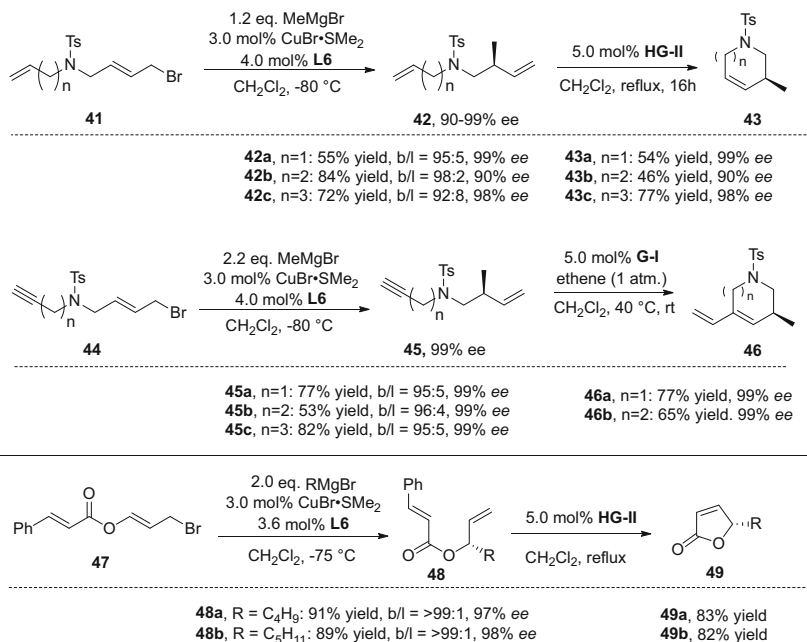
enantioselectivities were moderate (up to 60% ee). The obtained products were transformed to a variety of new phosphorus-containing chiral intermediates.

One major advantage of the allylic alkylation reactions is that it furnishes terminal olefinic groups, which represent ideal starting points for further modifications. One representative example is a tandem asymmetric allylic alkylation/intramolecular Heck reaction [43]. Copper–Taniaphos (**L6**) was used as catalyst to furnish chiral products **39** bearing a *o*-bromophenyl ether or amine in high regio- and enantioselectivity (up to 99% ee). The terminal olefin adjacent to the stereogenic centre formed was subjected to intramolecular Heck coupling, using a molten mixture of tetrabutylammonium bromide (TBAB) and tetrabutylammonium acetate (TBAA). The cyclised products **40** were obtained in excellent yields without racemisation at the stereogenic centre nor isomerisation of the exocyclic double bond (Scheme 17). The synthetic utility of these compounds was explored via the use of ring closing metathesis (RCM), stereoselective hydroborations, and reductions resulting in diverse synthetically relevant compounds.

Feringa and co-workers have shown that another interesting approach for the synthesis of N-heterocycles is combining Cu-catalysed AAA with RCM [44]. In this case, allylic bromides, substituted with a protected amine and terminal olefin (**41**) or alkyne (**42**) substituents, were alkylated with MeMgBr, and the obtained products **42** and **45** were subjected to RCM using Hoveyda–Grubbs (II) or Grubbs (I) catalysts. A collection of unsaturated piperidines, azepanes and azocanes were obtained in high yields and excellent enantioselectivities (Scheme 18, upper).

The same strategy was applied by this group for the preparation of γ -butyrolactone rings **49** [45]. In this case, the chiral olefin precursors **48** were prepared through their previously developed *hetero*-allylic asymmetric alkylation reaction [46] (*h*-AAA, where the allyl bromide contains a heteroatom at the gamma position) of allylic esters **47**, and Grignard reagents (Scheme 18, lower). The utility of this protocol was further demonstrated by converting these chiral lactones into the (–)-*whiskey* and (–)-*cognac lactone*. No loss of enantiomeric excess was observed in any of these transformations.

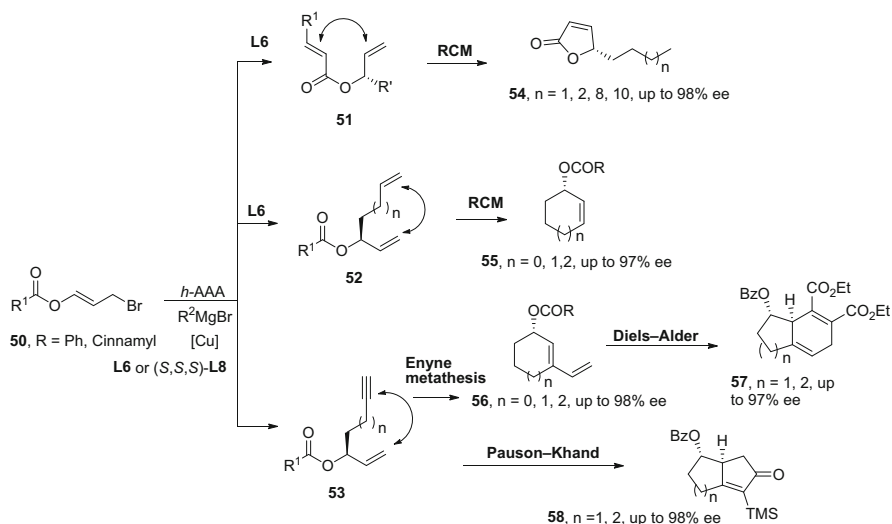
Combining the *h*-AAA reaction with RCM, the authors further reported on the preparation of a variety of chiral cyclic allylic esters **54–56** in high yields and



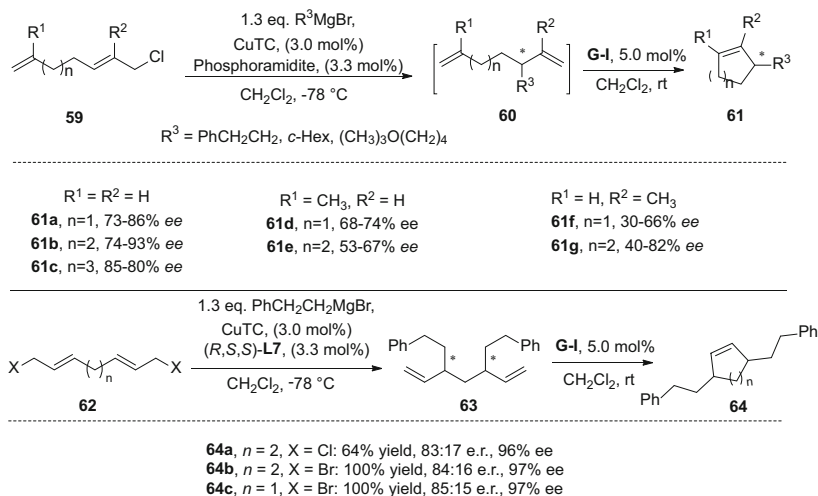
Scheme 18 Tandem asymmetric allylic alkylation/ring closing metathesis for the synthesis of N- and O-heterocycles

enantioselectivities (up to 98% ee) [47]. In addition, they also showed the diversity-oriented synthesis of different ring-fused bicyclic structures **57**, **58** through Diels–Alder reactions on 2,4-dienol esters **56** or Pauson–Khand reactions on enyne substrates **53** (Scheme 19).

Allylic alkylation in combination with RCM has also been employed by the group of Alexakis for the preparation of chiral carbocycles. In this case, ω -ethylenic allylic substrates **59** were subjected to Cu-catalysed AAA with primary and secondary Grignard reagents using phosphoramidite ligands [48, 49]. The choice of chiral ligand depends both on the length of the carbon chain between the allylic double bond and the ethylenic bond, as well as on the structure of the Grignard reagent. The subsequent RCM was performed in situ, without quenching the alkylation reaction, using Grubbs (I) catalyst. With simple unsubstituted ω -ethylenic allylic chlorides **59a–c**, five-, six- and seven-membered rings could be generated with enantioselectivities up to 86%, 93% and 85%, respectively. They also apply this strategy to similar substrates, bearing a methyl group on the terminal double bond (**59d,e**) or in the β -position (**59f,g**) of the allyl system, affording methyl-substituted five- and six-membered cycloalkenes **61d–g**, although with lower enantioselectivity (up to 82% ee). The authors anticipate metal coordination to the terminal olefin group prior to the σ -allyl complex formation to explain the variation on the enantioselectivity. Furthermore, the alkylation reaction on substrates with double allylic systems **62** was performed, which afforded compounds



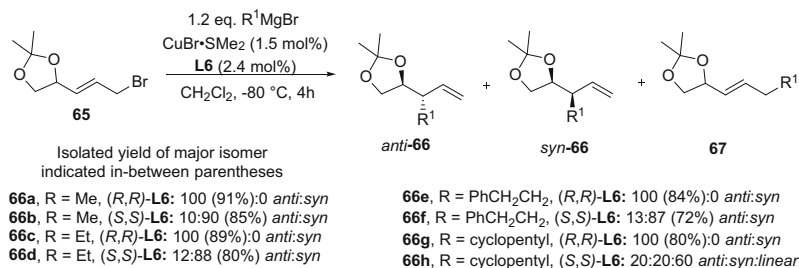
Scheme 19 Tandem asymmetric allylic alkylation/ring closing metathesis for the synthesis of chiral cyclic allylic esters and ring-fused bicyclic structures



Scheme 20 Construction of enantio-enriched carbocyclic compounds by asymmetric allylic alkylation and RCM

with two terminal vinyl groups **63** that were further subjected to intramolecular metathesis (Scheme 20).

Using both enantiomers of Taniaphos **L6** allowed Feringa and co-workers to perform the copper-catalysed AAA of δ -alkoxy-substituted allyl bromide **65** with high regio- and stereocontrol for both the *syn* and the *anti* isomers **66** [50]. When

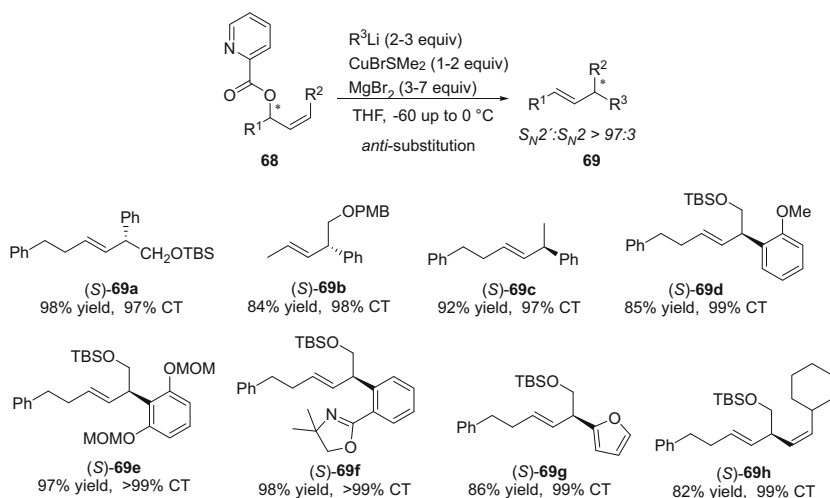


Scheme 21 Stereoselective synthesis of 1,2-hydroxyalkyl moieties **66** via Cu-catalysed AAA

allyl bromide **65**, which is readily obtained from commercial 1,2:5,6-di-*O*-isopropylidene-*D*-mannitol, was subjected to the reaction with MeMgBr using CuBr·SMe₂/*(R,R)*-Taniaphos (**L6**) as a catalyst, complete selectivity was observed towards the *anti* product **66**. The reaction with the (–)-enantiomer of the ligand led to the *syn* isomer with complete regioselectivity and high selectivity (*syn/anti* 90:10). Masking the alkoxy substituent in the dioxolane ring and performing the reaction at –80 °C are key factors to control the stereoselectivity of the reaction. A variety of Grignard reagents could also be used in the allylic alkylation with high selectivities in nearly all cases (Scheme 21). However, a strong mismatch effect was observed in the case of cyclopentyl magnesium bromide.

3 Asymmetric Allylic Substitutions Using Organolithium Reagents

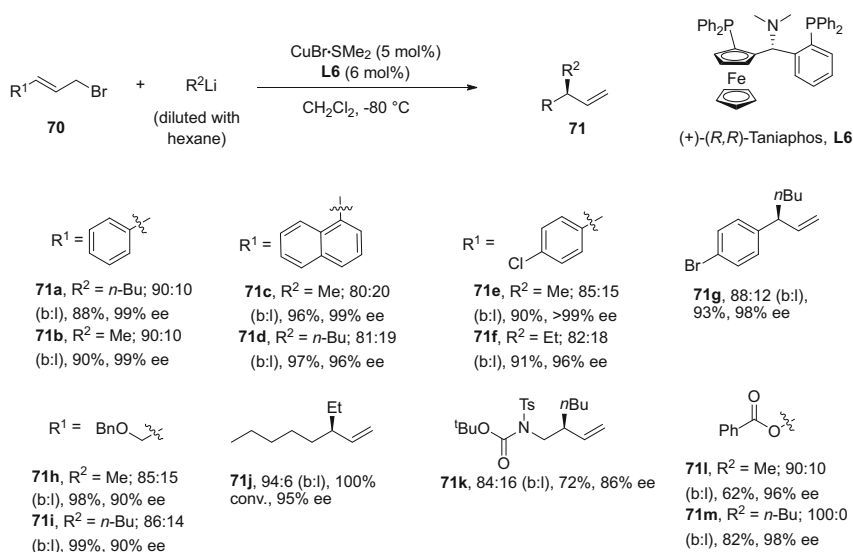
Recently, Feringa and co-workers reported for the first time the direct use of highly reactive organolithium reagents in Cu-catalysed AAA [51]. The application of these reagents in catalytic processes has been complicated due to their tendency to react spontaneously, by-passing the required involvement of the catalyst, but also due to their propensity to form aggregates in solution [52]. By the judicious choice of the chiral ligand and an exquisite control of the reaction conditions, the high reactivity of these organometallic reagents could be controlled affording the desired alkylated compounds with high stereochemical selectivity, without the presence of side products. More recently, the authors have reported novel highly efficient methodology for the palladium-catalysed direct cross-coupling of organolithium reagents with (hetero)aryl- and alkenyl(pseudo)halides under mild reaction conditions [53, 55] ([54] and references cited therein). However, a description of this palladium-based catalytic process is beyond the scope of this section. We will focus on the use of organolithium reagents in copper-catalysed asymmetric allylic substitution which includes the formation of tertiary and quaternary all-carbon stereogenic centres and their further application in the preparation of chiral building blocks for the synthesis of natural products and bioactive substances.



Scheme 22 Allylic substitution of optically active picolinates

In 2010, the group of Kobayashi described conditions for the asymmetric allylic substitution of chiral allylic picolinates **68** with aryl, alkenyl, and heteroaryl copper reagents in situ prepared from organolithium compounds and $CuBr \cdot SMe_2$ [56]. The reaction led to the *anti* S_N2' products **69** with high regioselectivity and chirality transfer (CT, defined by [% ee of product/% ee of starting material] $\times$ 100). The authors found that a number of experimental requirements were essential in order to obtain this high selectivity: (1) The addition of larger amounts of $MgBr_2$ with respect to Li^+ salts. (2) The use of 1–2 equivalents of $CuBr \cdot SMe_2$ with respect to the use of 2 equivalents of organolithium compound. (3) The geometry of the olefin in the starting material needs to be *cis*. The use of the corresponding *trans* picolinate still led to excellent selectivity although a *trans/cis* mixture of the olefin in the final product is obtained. Different allylic picolinates **68** were tolerated in the substitution reaction (R^1 and $R^2 = c\text{-Hex}$, $PMBOCH_2$, $TBSOCH_2$, Me , $Ph(CH_2)_2$, $PMBO(CH_2)_3$) as well as a wide variety of organolithium compounds, which were prepared by Li-halogen exchange or direct *ortho* metalation. Interestingly, *Z*- and *E*-alkenyl lithium could also be employed providing the *anti* S_N2' product with high regioselectivity and CT, with retention of the olefin geometry (Scheme 22).

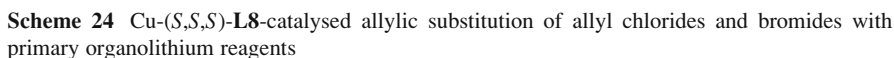
In 2011, the group of Feringa reported that the use of catalytic amounts of in situ formed chiral copper complexes, generated from copper salts and chiral phosphorus-based ligands in CH_2Cl_2 as solvent, is an effective way to control the high reactivity of alkyllithium compounds allowing for an efficient and highly selective asymmetric allylic alkylation process [51]. Different factors, typically associated with the use of organolithium compounds, were required to be considered in order to achieve a highly selective and high-yielding reaction. (1) A fast lithium-leaving group exchange has to be suppressed by efficient transmetalation to the Cu complex. (2) After oxidative addition of the allylic system, a fast

**Scheme 23** Cu-catalysed AAA with alkyl lithium compounds

reductive elimination from the σ -allylcopper complex, leading to the S_N2' product, should also be promoted. If the reductive elimination step is slow, the undesired linear product can be formed via isomerisation from the Cu(III) σ -complex to the Cu(III) π -allyl complex, followed by reductive elimination [17]. (3) Finally, if the substrate contains an aryl halide, lithium–halogen exchange would lead to the formation of aryl lithium species which can give a new range of undesired side products.

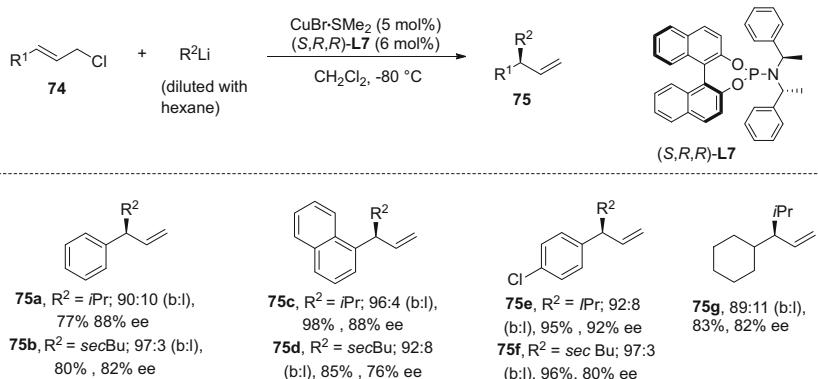
The study of the reaction between n -BuLi, the most well-known organolithium reagent, and cinnamyl bromide **70a** was chosen to establish optimised conditions for the AAA reaction. During initial experiments, the authors observed that the use of non-etheral solvents maintains the aggregation of the organolithium species within the reaction prior to transmetalation to copper, which in turn reduces its intrinsic reactivity and thereby, as a consequence, avoids a plethora of side reactions. Among the different ferrocenyl-type chiral diphosphines ligands screened, (*R,R*)-(+)-Taniaphos **L6** at $-80^\circ C$ gave the best results affording the branch product **71** in high S_N2'/S_N2 (83:17) regioselectivity and enantioselectivity (98% ee). In order to achieve these excellent selectivities, the organolithium reagent needs to be diluted with n -hexane and added to a solution of the catalyst and allylic bromide in dichloromethane at $-80^\circ C$. Methyllithium is an exception where toluene was used for dilution and addition, as the use of n -hexane as co-solvent gave solubility problems with this organolithium reagent.

Under the optimised reaction conditions, different alkyl chains in the lithium reagent were tolerated providing excellent regio- and enantioselectivities (Scheme 23). Importantly, the reaction tolerates the presence of aryl halide

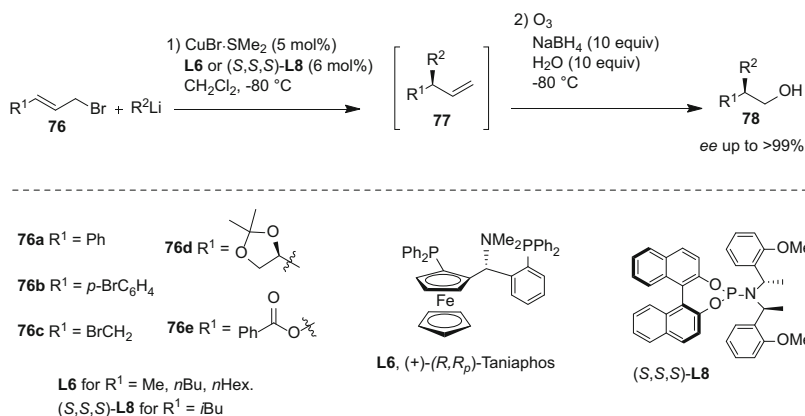


Interestingly, the reaction also tolerates various functional groups known to be sensitive to organolithium compounds. These include benzyloxy- and *N*-Boc-protected amine groups, esters and alcohols.

The use of secondary lithium reagents constitutes a second limitation for the CuBr-SMe₂/Taniaphos system [57]. Phosphoramidites were again the most effective ligands, in combination with allyl chlorides, for this transformation, and the use of a CuBr-SMe₂/(*S,R,R*)-**L7** catalyst system was found to be optimal for the addition of *sec*-BuLi and *i*-PrLi. Interestingly, under these conditions the use of



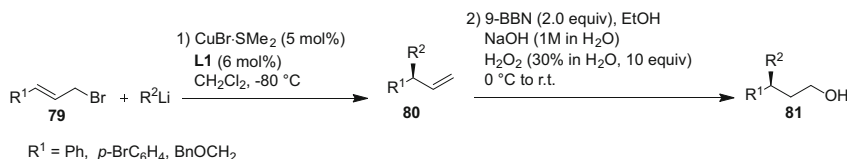
Scheme 25 Cu/L4-catalysed allylic substitution of allyl chlorides and bromides with secondary organolithium reagents



Scheme 26 One-pot synthesis of β-alkyl-substituted alcohols **27** through Cu-catalysed asymmetric allylic alkylation of allyl bromides with organolithium reagents followed by ozonolysis and reduction

the corresponding allyl bromides led to a decrease in selectivity as observed for the reaction between cinnamyl bromide and *sec*-BuLi (S_N2'/S_N2 79:21, 38% ee). As reported before, this catalyst system tolerates sterically hindered 1-naphthyl-substituted allyl chlorides (**75c-d**) or halogen-containing cinnamyl substituents (**75e-f**), although lower selectivity was observed for the reaction with cyclohexyl-substituted allyl chloride **75g** (Scheme 25). In general, the transfer of the *sec*-butyl chain proceeds with slightly lower enantioselectivity (up to 82% ee).

As mentioned above, one of the advantages of the AAA reaction is that it gives access to products with a carbon stereogenic centre next to a terminal olefin, which can be further functionalised. Taking advantage of this structural feature, Feringa and co-workers developed a one-pot synthesis of β- and γ-alkyl-substituted alcohols



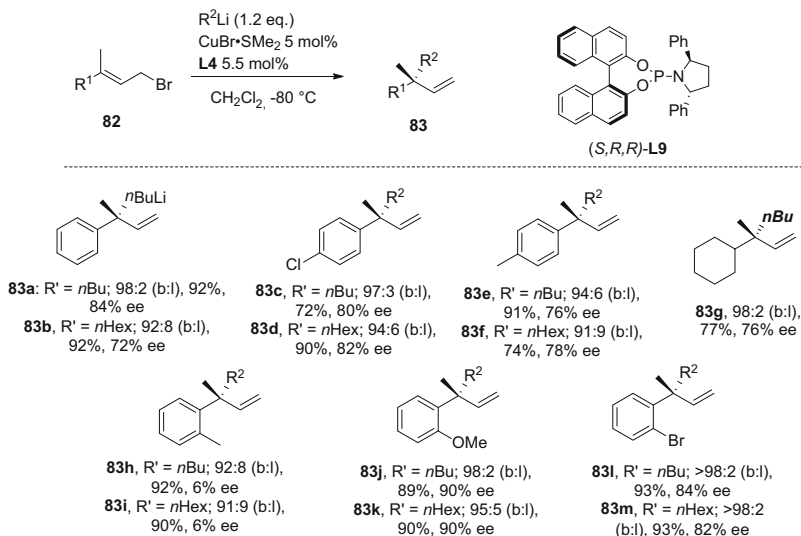
Scheme 27 Synthesis of γ -alkyl-substituted alcohols through Cu-catalysed asymmetric allylic alkylation of allyl bromides with organolithium reagents followed by a hydroboration/oxidation

(**78** and **81**, respectively) (Schemes 26 and 27) through a tandem copper-catalysed asymmetric allylic alkylation (AAA) with organolithium reagents followed by reductive ozonolysis or hydroboration/oxidation, respectively [59]. In the first case, after Cu-catalysed AAA, using **L7** or (*S,S,S*)-**L8** as ligand, and ozonolysis of the corresponding alkene **77**, the use of ten equivalents of NaBH_4 and ten equivalents of water was necessary to achieve full conversion of the desired β -alkyl-substituted alcohols **78**. This one-pot protocol was applied to a variety of lithium reagents such as MeLi , *n*- BuLi and *n*- HexLi and different allylic bromides counterparts, bearing aromatic (**76a,b**) and aliphatic substituents (**76c,d**), as well as an ester functionality (**76e**), generating the corresponding alcohols with excellent ee's (72–99%) and good overall yields (up to 90%) (Scheme 26).

The corresponding γ -alkylated alcohols **81** were also obtained in good yields (67–93%), without erosion of the enantiomeric excess, when the terminal olefins **80** were subjected to hydroboration/oxidation reaction using 9-BBN (Scheme 27).

To achieve catalytic enantioselective formation of all-carbon quaternary stereogenic centres, a carbon atom bound to four distinct carbon substituents is a considerable synthetic challenge in organic synthesis [60, 61]. In the case of acyclic systems, it is even more complicated due to the number of degrees of freedom associated with these structures. In the past decades, impressive progress has been made in this field, and several synthetic transformations, including enantioselective allylic substitution and conjugate additions, enantioselective alkylation, nucleophilic allylations, aldol reactions, Mannich reactions, rearrangements reactions and catalytic and enantioselective intermolecular Heck-type reactions, have been described for the construction of these highly congested structural moieties [62, 63].

As part of their research programme on the direct use of organolithium reagents in AAA, Feringa and co-workers developed methodologies for the synthesis of all-carbon quaternary stereogenic centres via copper-catalysed AAA of *E*- and *Z*-trisubstituted allyl bromides [57]. While examining the reaction between *n*- BuLi and an *E*-trisubstituted allyl bromide **82a** and after extensive screening of different phosphoramidite ligands, it was found that chiral ligand **L9** was optimal for the reaction, providing product **83a** in high regioselectivity ($\text{S}_{\text{N}}2'/\text{S}_{\text{N}}2$ 98:2) and good enantioselectivity (84% ee, Scheme 28). Different primary alkyl organolithium reagents gave excellent regioselectivity (up to >98:2) and moderate to good enantioselectivities (ranging from 72% to 84% ee) although the use of secondary alkyl lithium compounds was not shown. *p*-Methyl- and *p*-chloro-substituted

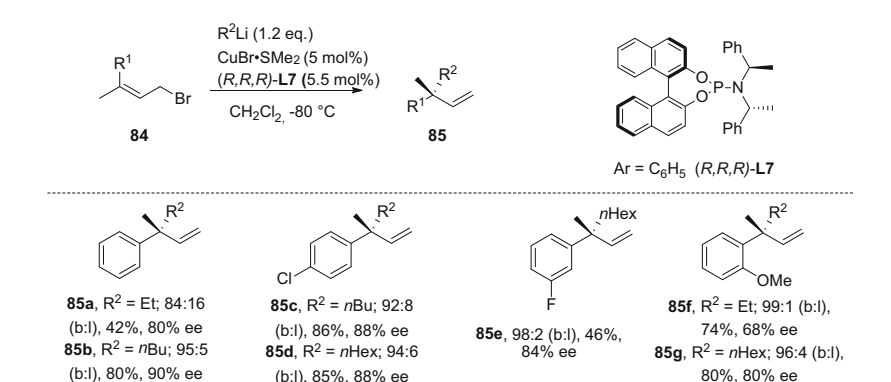


Scheme 28 Enantioselective Cu/**L15**-catalyzed synthesis of quaternary carbons with organolithium reagents

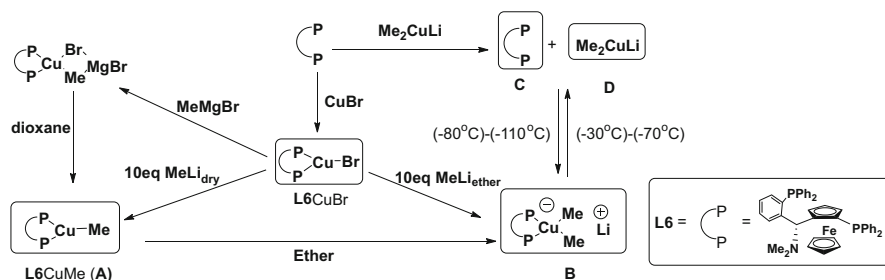
cinnamyl bromides (**82c–f**) were found to be effective electrophilic partners, as well as a cyclohexyl-substituted derivate **82g**.

Interestingly, it was observed that the nature of the *ortho* substituents at the aryl ring plays an important role in the stereochemical outcome of the AAA reaction. The use of simple methyl groups (**83h–i**) led to nearly racemic products, independently of the nature of the organolithium compound used. However, the enantioselectivity was enhanced, when the corresponding *o*-bromo- or *o*-methoxy-substituents (**83j–m**) were used instead. The authors suggest a coordination between these *ortho* substituents and the copper complex as a key parameter in the stereo-discrimination event that allows to achieve these high enantioselectivities (Scheme 28).

In a subsequent report [64], Feringa and co-workers studied the effect of the geometry of the prochiral trisubstituted allyl substrate on the stereochemical outcome of the copper-catalysed AAA with organolithium reagents. Various *Z*-trisubstituted allyl bromides **84** were examined, and it was found that phosphoramidite (*S,R,R*)-**L9** was the most effective ligand, affording the corresponding all-carbon quaternary stereogenic centres in good enantioselectivities (Scheme 29). As before (Scheme 28), different organolithium reagents including EtLi, *n*-BuLi and *n*-HexLi were tolerated providing good regio- and enantioselectivities (up to 90% ee) (Scheme 29). The substrate scope includes *p*-chloro-, *m*-chloro- and *m*-fluoro-substituted allyl bromides and does not suffer from parasitic lithium–halogen exchange. In general, slightly higher enantiomeric ratios were obtained with *Z*-allyl bromides **84** when using this catalytic system (CuBr·SMe₂/(*S,R,R*)-**L7**) than for the previously reported *E*-allyl bromides (using CuBr·SMe₂/(*S,R,R*)-**L9**) [57]. An exception is the use of *o*-methoxy-substituted allyl bromide (**84f**) where a



Scheme 29 Enantioselective Cu-(*R,R,R*)-**L7**-catalysed synthesis of quaternary carbons with organolithium reagents



Scheme 30 Copper species formed from alkylmagnesium and alkyllithium compounds in the AAA

decrease in the enantiomeric ratio was observed. In contrast with the results obtained by the groups of Hoveyda [65] and Sawamura [66] but in agreement with a previous findings of the Feringa group [42], this catalyst system led to the same enantiomer independently of the use of the E and Z geometry of the trisubstituted allyl bromide. This observation suggests lack of interaction between the copper catalyst and the leaving group of the substrate prior to the oxidative addition taking place during the catalytic cycle.

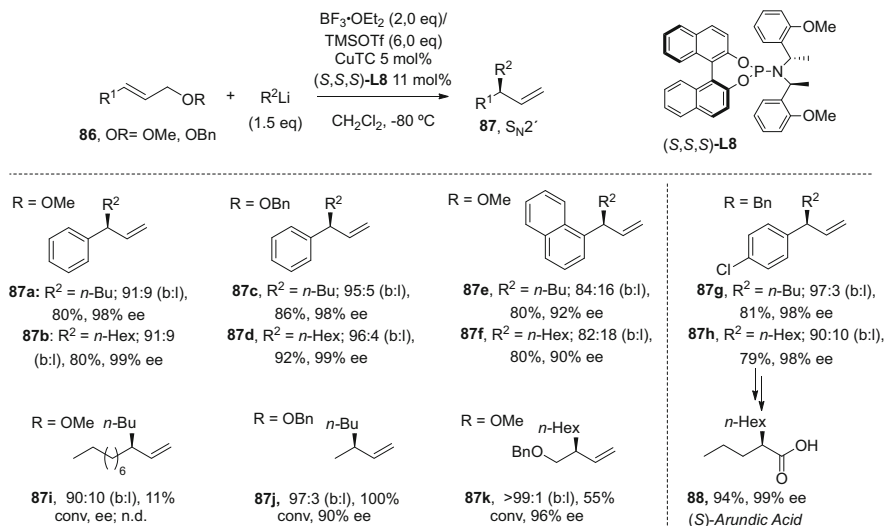
Mechanistic studies were also performed to elucidate the nature of the chiral copper complex which is responsible for the high activity and selectivity in the AAA and to elucidate the dramatic effect of the solvent on the selectivity of the reaction [51, 57]. The effect of different combinations of copper salts, solvents and quantities of MeLi over a range of temperatures by ^1H , ^{31}P and $^6\text{Li}/^7\text{Li}$ NMR spectroscopy was examined. Similarly to the corresponding Grignard reagents [17, 67], the authors suggested that the first step in the AAA catalytic cycle is the formation of the chiral methyl copper complex, via transmetallation between the methyllithium compound and the ligand-bound copper species (**L6CuBr**). Indeed, by employing an excess of ten equivalents of MeLi and copper complex **L6CuBr** in

CH_2Cl_2 at -80°C , the exclusive formation of **L6CuMe** was observed (complex **A**, Scheme 30). This copper species was independently confirmed via the reaction between **L6CuBr** complex and the corresponding Grignard reagent (MeMgBr or Me_2Mg) followed by removal of MgBr_2 by coordination with dioxane [68]. Additionally, the use of stoichiometric amounts of this copper species (**L6CuMe**, **A**) in the AAA of cinnamyl bromide led to the alkylation product with high enantiomeric excess ($>98\%$ ee).

Remarkably, lithium does not seem to have any interaction with the active catalyst, unlike with the corresponding Grignard reagents where Mg is part of the catalytically active copper species. Indeed, in the absence of Et_2O , the enantioselectivity of the reaction is independent of the nature of copper source used. When Et_2O was added to the solution, the mixture resulted in the formation of fragile bimetallic **B** that is in a temperature-dependent dynamic equilibrium with the non-ligated organo-cuprate LiCuMe_2 **D**. The low enantioselectivity observed (28% ee) when the organolithium compound is diluted (Et_2O) supports the existence of non-ligand-bound cuprate **D**.

Allylic halides are very useful and versatile substrates for the Cu-catalysed AAA with organometallic reagents, with many applications in the synthesis of natural product and pharmaceuticals. Nonetheless, the use of these reactive electrophilic partners could constitute a drawback when the alkylation reaction needs to be performed at the latest stages of a multistep synthesis. A possible solution would consist in the use of more robust and inert leaving groups in the allylic substrate that can initially act as a protecting group along the synthetic sequence, but when desired, it can be activated to operate as a leaving group for the asymmetric alkylation reaction. With this aim, the use of stable alkyl- and aryl-protected allylic ethers as electrophilic partners in Cu-catalysed AAA with organolithium reagents was evaluated [69].

Screening of different allylic esters and ethers in the reaction with *n*-BuLi, in the presence of different Lewis acids for the activation of the allyl ether, revealed that the use of $\text{CuTC}/(S,S,S)\text{-L8}$, together with a mixture of $\text{BF}_3\cdot\text{OEt}_2$ and TMSOTf, as optimal combination. The use of the $\text{BF}_3\cdot\text{OEt}_2/\text{TMSOTf}$ mixture, which in situ forms the new Lewis acid BF_2OTf , was found to be indispensable to obtain high selectivity. The reaction does not proceed in the absence of $\text{BF}_3\cdot\text{OEt}_2$, thus allowing the ether to act as a robust protecting group. Under these conditions, similar regio- and enantioselectivities, compared to the previous results, using allyl halides were obtained. High regio- and enantioselectivities were obtained for the reaction between *n*-BuLi and *n*-HexLi with linear aliphatic and aromatic substrates **86**, including the use of hindered 1-naphthyl substituents as well as aromatic chlorides (Scheme 31). An interesting observation was made for a substrate bearing both an allylic -OMe and -OBn ether **86k**, where substitution occurred preferentially towards the -OMe group rather than the -OBn group. The synthetic utility of this novel Cu-catalysed AAA was demonstrated in the three-step asymmetric synthesis of biologically active (*S*)-arundic acid **88**, with high enantioselectivity ($>99:1$ e.r., Scheme 31).



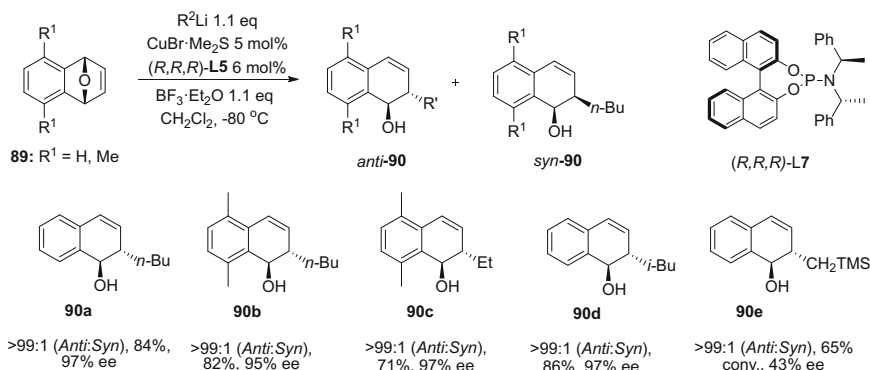
Scheme 31 Copper-catalysed AAA of acyclic allylic ethers with organolithium reagents

The catalytic enantioselective ring opening of oxabicyclic alkenes by carbon-based nucleophiles is a useful method for the simultaneous introduction of multiple stereocentres under high stereocontrol ([70] and references cited therein). Recently, several methods have been developed for ring opening of these *meso* heterobicyclic compounds employing different soft- and organometallic nucleophiles [71, 72].

In general, the use of transition-metal catalysts led to the *syn* product, as a result of the *exo* attack of the nucleophile to the oxabicyclic alkene. Of particular interest is the palladium-based methodology using organozinc reagents, developed by the group of Lautens, which proceeds with high selectivity for the *syn* diastereomer [73]. Exceptions to the formation of the *syn* ring-opened products, leading to high *anti* selectivity, have been reported using copper-based catalysts. In 2002, an unprecedented copper/phosphoramidite-catalysed enantioselective alkylative ring opening of oxabenzonorbornadiene derivatives with dialkylzinc reagents was reported by Minnaard and Feringa [74]. This transformation reaction shows a high level of *anti* stereoselectivity (up to >99:1 *anti/syn*). In 2003, the group of Carretero also reported conditions for the copper-catalysed *anti* stereoselective ring opening of oxabicyclic alkenes with Grignard reagents [75].

In 2012, the group of Feringa described the use of organolithium reagents as nucleophiles for the asymmetric ring opening of oxabicyclic alkenes, with excellent *anti* selectivity and enantioselectivity [76].

Screening different reaction parameters in the asymmetric ring opening of compound **89a** with *n*-BuLi, it was found that the use of a copper/phosphoramidite (*R,R,R*)-**L7** complex, together with a Lewis acid (BF₃·OEt₂) at −80 °C, allowed for the exclusive formation of the *anti* diastereoisomer with excellent



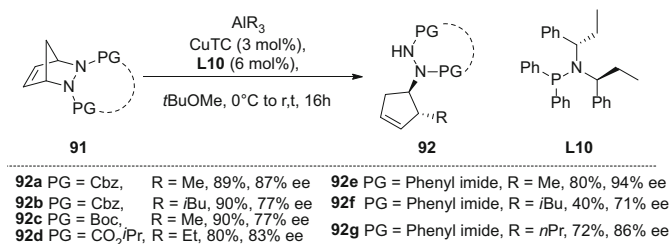
Scheme 32 Cu-catalysed asymmetric ring opening of oxabicyclic substrate **89** with organolithium reagents

enantioselectivity (97% ee). The reaction tolerates the use of primary (*n*-BuLi, *n*-HexLi, EtLi) and secondary organolithium compounds (*i*-BuLi) providing in all cases full conversion and high selectivity towards the desired *anti* products **90**. (Scheme 32). Remarkably, the bifunctionalised organolithium reagent, TMSCH₂Li, was used for the first time in a catalytic asymmetric process although decreased reactivity and enantioselectivity were observed (**90e**, 43% ee).

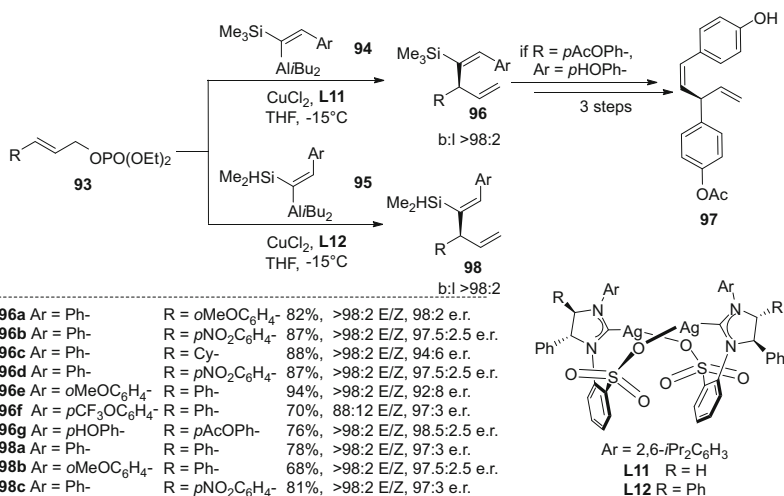
4 Asymmetric Allylic Substitutions Using Organoaluminium, Zinc and Zirconium Reagents

In 2010, Alexakis and co-workers reported the enantioselective ring opening desymmetrisation of meso-hydrazines with trialkylaluminium reagents catalysed by copper (Scheme 33) [77]. This process is proposed to follow a classical allylic substitution pathway, with the organoaluminium reagent playing an important role in the activation of the leaving group taking advantage of its Lewis acidity, in addition to being an alkyl donor. After testing several ligands, SimplePhos **L10** proved to be optimal allowing for various organoaluminium species to selectively add onto *N*-carbamate-protected meso-hydrazines **91** in good yields (up to 90%) and selectivities (94% ee). Phosphoramidites were also tested as ligands but, in some cases, cleaved to the corresponding phosphine amines, resulting in poor selectivities as a consequence.

In 2010, Hoveyda and co-workers tackled the problem of the stereo- and enantioselective synthesis of chiral 1,4-dienes, using allylic substitution, and developed an efficient *one-pot* protocol to access these valuable intermediates [78]. They showed that by reacting vinylaluminium reagents with allylic phosphates **93** in the presence of a copper–NHC catalyst **L11–12**, it was possible to obtain both geometrically and enantiomerically pure 1,4-dienes in excellent yields. This is



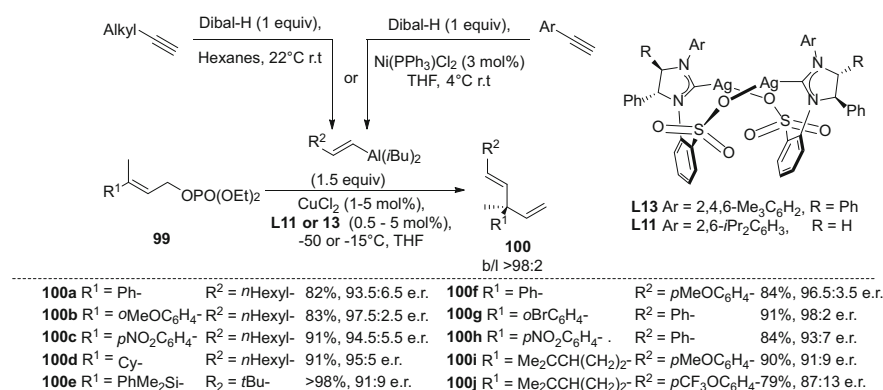
Scheme 33 Copper-catalysed ring opening of meso-hydrazines



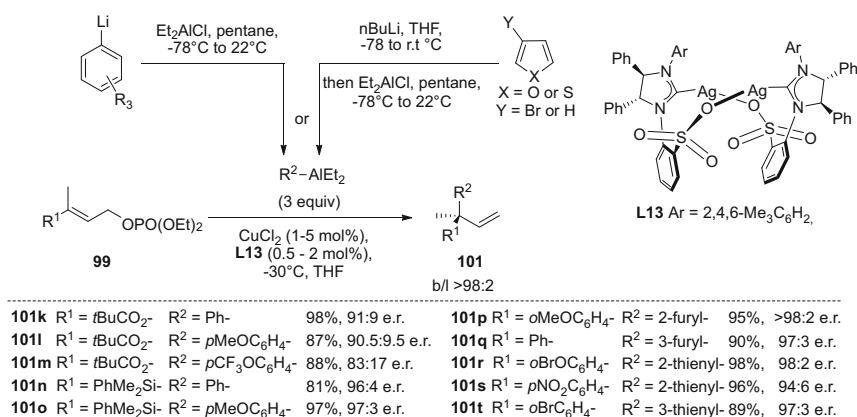
Scheme 34 Copper-catalysed enantioselective synthesis of 1,4-dienes containing Z or E alkenes

achieved by carefully controlling the *E/Z* geometry of the starting vinylaluminium reagent as the olefin's stereo information is transferred without erosion to the final product during this reaction (Scheme 34). Both electron-rich and electron-poor allylic phosphates perform well in combination with diverse aromatic vinyl aluminium reagents (Scheme 34, entries **96a-g**, **98a-c**). Further exploring this method, they were able to achieve the first enantioselective synthesis of *nyasol* **97** (Scheme 34, upper, **96g**).

As a follow-up of the aforementioned studies, Hoveyda and co-workers then addressed one of the limitations in scope their method still had. Indeed, more sterically demanding Si-containing aryl vinyl aluminiums were incompatible with the reaction conditions. Hence, they turned their attention to alternative methods for the preparation of organoaluminiums compatible with the asymmetric allylic substitution of allylic phosphates for the formation of quaternary stereogenic centres. Firstly, they studied the enantioselective substitution of alkyl vinyl aluminiums obtained from the hydroalumination with DIBAL-H of alkyl-substituted alkynes (Scheme 35) [65]. When the proper catalyst is chosen (**L11** as chiral ligand), this

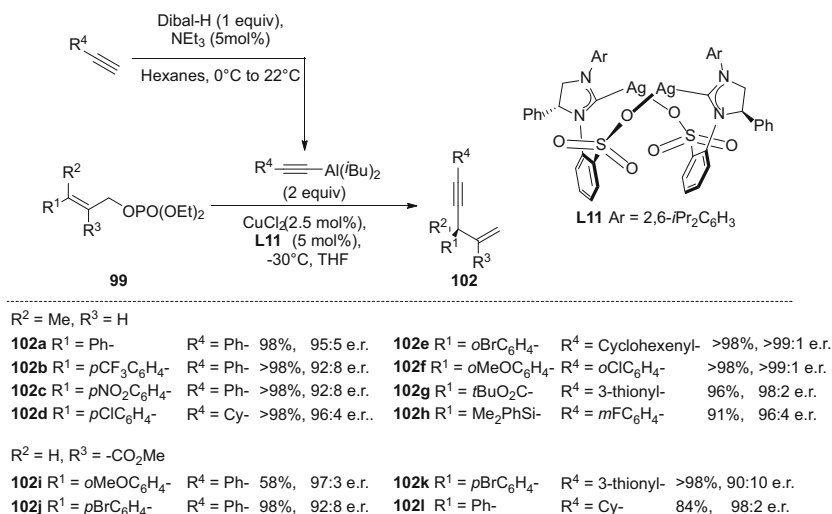


Scheme 35 Copper-catalysed enantioselective synthesis of 1,4-dienes via hydroalumination



Scheme 36 Copper-catalysed enantioselective synthesis of diaryl bearing quaternary carbon stereogenic centres

reaction proceeds to provide good yields and high regio- and enantioselectivities with a wide range of substrates (Scheme 35, entries **100a-e**). Secondly, they focused on aryl-substituted organoaluminium reagents. These proved more challenging, as a non-negligible amount of alkynyl addition product was formed under standard conditions. This most probably results from the deprotonation of the corresponding alkyne to afford the alkynylaluminium which subsequently reacts with the allyl phosphate. This issue was solved by the addition of a catalytic amount of Ni (PPh₃)₂Cl₂ during the hydroalumination step, which significantly enhanced the rate of the desired reaction, thereby eliminating the formation of the by-product. This mixture, when added to the allyl phosphate in the presence of copper chloride and **L13**, gave the allylic substitution product with good yields and high regio- and enantioselectivities with a range of substrates (Scheme 35, entries **100f-j**).

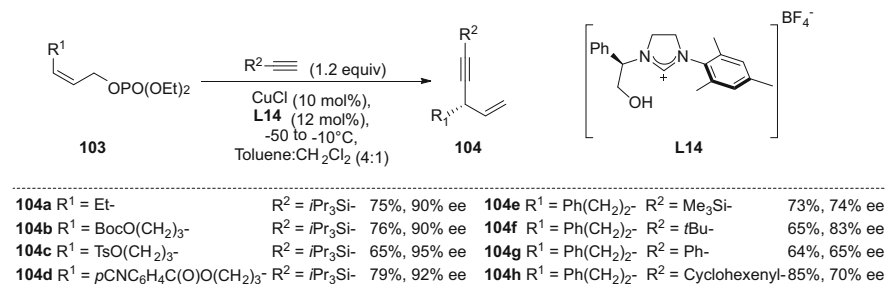


Scheme 37 Copper-catalysed enantioselective synthesis of alkyne-substituted stereogenic centres

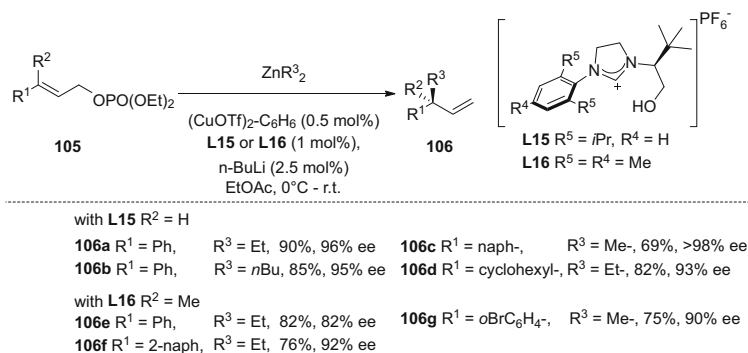
In parallel Hoveyda and co-workers also studied the asymmetric allylic substitution of arylaluminium species on similar allyl phosphates to generate valuable and difficult to obtain diaryl compounds bearing chiral quaternary carbon stereocentres (Scheme 36) [65, 77–79]. By reacting aryllithium species with diethylaluminium chloride, followed by direct addition to an allyl phosphate in conjunction with copper and N-heterocyclic carbene ligand **L13**, quaternary stereocentres were obtained in excellent yields and good enantiomeric ratios for both aromatic and heteroaromatic compounds.

Hoveyda and co-workers also exploited the reactivity of the alkynyl aluminium compounds for the formation of alkyne-substituted stereogenic centres, which were initially observed as a by-product during their study of the synthesis of 1,4-dienes via hydroalumination (Scheme 35) [73]. By altering the previous hydroalumination protocol and adding a catalytic amount of triethylamine instead of a nickel catalyst to the reaction mixture, they could switch the selectivity of that step to give solely the alkynylaluminium intermediate. These could then be used under similar conditions to give the corresponding alkyne-substituted enantiopure quaternary stereogenic centres with various substrates (Scheme 37, entries **102a–h**) [37]. Subsequently it was also shown that ester-substituted allyl phosphates could also serve as substrates for this reaction (Scheme 37, entries **102i–l**) [80].

Recently Ohmiya and Sawamura reported a similar transformation based on in situ formed lithium acetylides [81]. With the perspective of expanding the scope of existing methods by avoiding the need for organoaluminium species, which had a strong impact on the functional group tolerance of the reaction, they investigated the possibility of using lithium acetylides obtained by deprotonation of terminal alkynes in combination with (Z)-allyl phosphates and a copper N-heterocyclic



Scheme 38 Copper-catalysed enantioselective allylic alkylation of terminal alkynes



Scheme 39 Hydroxyalkyl NHC copper-catalysed enantioselective allylic alkylation on allylic diethyl phosphates

carbene catalytic system (Scheme 38). Mechanistical investigations showed that the presence of the coordinating phenol group on the ligand (L14) was crucial to obtain high enantioselectivities. First, it avoided ligand dissociation, which had proven problematic, and second it allowed for a tighter bridged transition state between the counteranion, acetylide carbon and phosphate leaving group resulting in a regio- and enantiodiscriminating step and showing higher selectivity. The geometry of the allyl double bond proved also to be crucial as (*E*)-allyl phosphates gave poor yields and enantioselectivities.

Mauduit and co-workers explored the effect of various ligands on copper-catalysed enantioselective allylic alkylation reactions and showed that chelating hydroxyalkyl NHCs were efficient ligands for the asymmetric alkylation of organozinc reagents to allylic diethyl phosphates (Scheme 39, entries 106a-d) [82]. After a screening of their ligands towards this reaction in several solvents, it became apparent that a combination of copper(I) triflate and ligand L15 in ethyl acetate at room temperature afforded good yields and selectivities for this transformation, even at low catalyst loadings. Subsequently the use of a novel family of NHC's was reported for the reaction between organozinc reagents and 3-methyl-

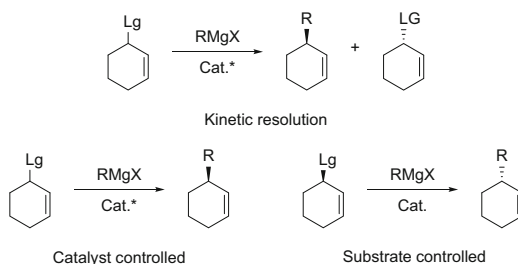
allyl phosphates to generate quaternary centres under the same conditions, with **L16** providing the best results (Scheme 39, entries **106e-g**) [83].

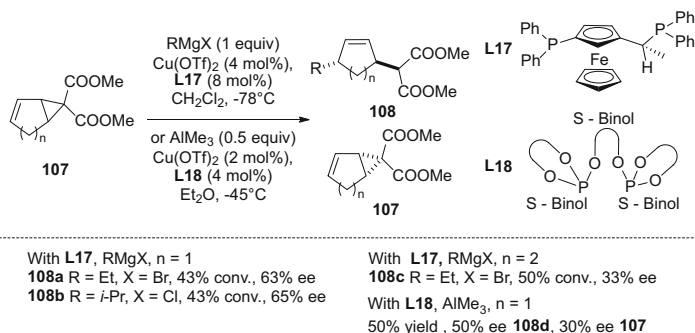
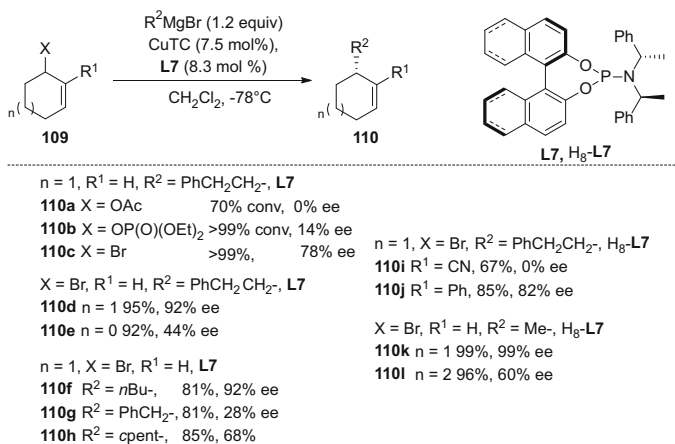
5 Asymmetric Allylic Substitutions on Racemic Substrates with Organometallic Reagents

In addition to the previously described examples of copper-catalysed asymmetric allylic alkylation on non-chiral substrates using organometallic reagents, some reports, dedicated to performing the same reaction on chiral compounds, have appeared over the past few years. Carrying out allylic substitution on this type of substrates remains a great challenge due to the additional and distinct diastereomeric interactions existing between the catalyst and each enantiomer, in comparison to non-chiral substrates. These different interactions, and therefore reactivity, have been taken as an advantage to perform kinetic resolutions on certain racemic allylic substrates [84]. However, the major limitation of simple kinetic resolution, a maximum theoretical yield of 50%, has led to a shift in focus to other, novel, mechanistic concepts which, in theory, allow for the quantitative allylic alkylation of racemic compounds [37, 65, 77–86]. These revolve around two major principles, the first involves constructing a catalytic cycle containing a step in which the chiral information in the starting substrate is erased, followed by a step in which new chiral information is introduced. This was done either by inducing rapid interconversion between the two starting enantiomers, with only one being reactive (dynamic kinetic resolution), or by developing a system which is able to pass via either a C2 symmetrical or a meso transition state (dynamic kinetic asymmetric transformation). A second approach involves establishing two enantioconvergent pathways from each enantiomer to the same product (for one of the first examples, see [87]). Reactions where the configuration of the starting substrate dictates the stereochemical outcome of the reaction, called enantiospecific allylic alkylations, are not pertinent to this chapter and will therefore not be treated here (Scheme 40) (for a review covering these, see [88]).

Alexakis and co-workers employed copper in conjunction with a variety of ligands to perform the kinetic resolution of cyclic alkenyl cyclopropane malonates using trimethylaluminium or alkylmagnesium reagents in moderate yields and

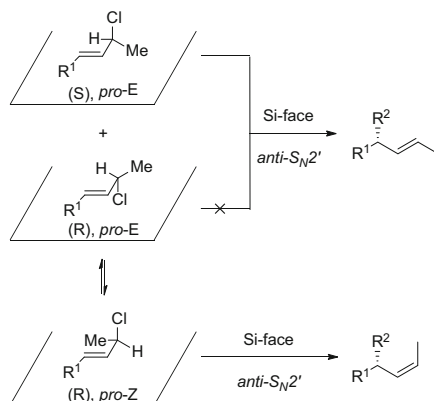
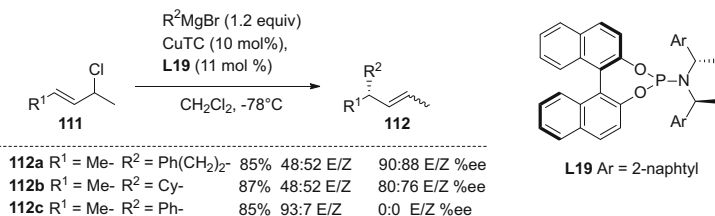
Scheme 40 Substrate-controlled vs. catalyst-controlled allylic alkylation on chiral substrates



**Scheme 41** Kinetic resolution of cyclic alkenyl cyclopropane malonates**Scheme 42** Copper-catalysed asymmetric allylic alkylation of racemic cyclic substrates

selectivities (Scheme 41) [89]. Of the diverse ligands tested, ferrocenyl-based **L17** gave the best overall results. Under these conditions, organozinc as well as arylmagnesium reagents failed to react.

In the same year Alexakis and co-workers also described the copper-catalysed allylic alkylation of organomagnesium reagents onto racemic cyclic substrates **109**. They based themselves on the idea that a dynamic kinetic asymmetric transformation could be employed to reach an enantiopure product starting from a mixture of both enantiomers [90]. After extensive screening, it was shown that phosphoramidite ligands gave the best results and that the outcome of the reaction was strongly dependent on both the solvent and leaving group used (Scheme 42, **110a-c**). Preliminary mechanistical investigations suggested that the outcome of the reaction was indeed governed by a dynamic kinetic asymmetric transformation. In terms of scope, a wide range of organomagnesium reagents were tested to provide good yields and moderate to good enantioselectivities (Scheme 42, **110f-h**,

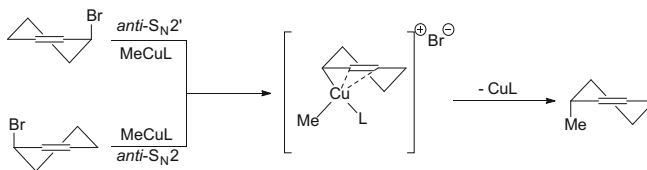
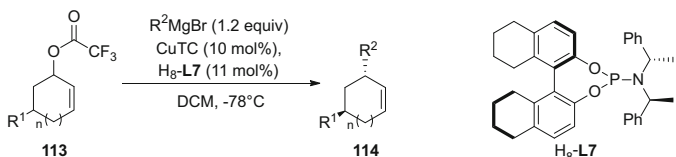


Scheme 43 Copper-catalysed asymmetric allylic alkylation of racemic substrates

110k-l). The reaction also tolerates cyclic substrates of diverse sizes and substitution patterns, though only 3-bromocyclohexanes gave good results, while larger rings yield only moderate selectivities (Scheme 42, **110d-e**, **110i-l**).

In a follow-up study, Alexakis and co-workers investigated whether their dynamic kinetic asymmetric transformation could be applied to non-cyclic substrates [91]. Here, it was observed, after some optimisation, that (*E*)-4-chloropent-2-enes **111** smoothly reacted with an organomagnesium reagent to give a mixture of enantio-enriched E/Z-alkylated products **112** with the same absolute configuration for both isomers. Based on experimental observations, the authors postulate that the catalyst, due to substrate–ligand interactions with the vinylic methyl group (R¹ = Me), will react in an *anti*-fashion, specifically with the Si face of the substrate, and, hence, force one enantiomer to take a pro-Z conformation before reacting (Scheme 43). This behaviour, named stereodivergent kinetic resolution, was consistent with several alkylmagnesium species, but failed when arylmagnesiums were used.

In 2012 Alexakis and co-workers, prompted by the uncertainties remaining in their mechanistic investigations of their allylic alkylation of racemic cyclic substrates, extended their study of the mechanism of the reaction by turning to computational methods and deuterium-labelling experiments [92]. Modelling of the potential transition states, with emphasis on the interaction of each enantiomer with the ligand, pointed towards the fact that, rather than undergoing a dynamic

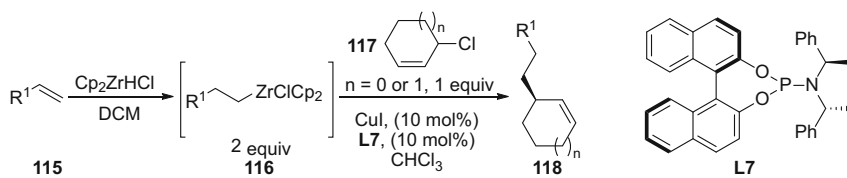
**Scheme 44** Enantioconvergent pathway proposed by Alexakis and co-workers

114a $n = 1, R^1 = H, R^2 = Ph(CH_2)_2-$ 85%, 97% ee

114b $n = 1, R^1 = H, R^2 = \text{cpent-}$ 86%, 90% ee

114c $n = 1, R^1 = Me, R^2 = Ph(CH_2)_2-$ 80%, 93% ee

114d $n = 0, R^1 = H, R^2 = Ph(CH_2)_2-$ 78%, 62% ee

Scheme 45 Copper-catalysed asymmetric allylic alkylation of racemic cyclic substrates with trifluoroacetate as leaving group

118a $n = 1, R^1 = nBu-$ 56%, 95% ee **118d** $n = 1, R^1 = pMeOC_6H_4-$ 63%, 95% ee

118b $n = 1, R^1 = TMS-$ 52%, 93% ee **118e** $n = 0, R^1 = Ph(CH_2)_2-$ 57%, 92% ee

118c $n = 1, R^1 = oBrC_6H_4-$ 87%, 87% ee **118f** $n = 0, R^1 = Br(CH_2)_9-$ 42%, 92% ee

Scheme 46 Dynamic kinetic asymmetric allylic alkylation of 3-substituted cyclohexenes

kinetic resolution based on a common meso- π -allyl intermediate, the reaction proceeds via two divergent pathways for each enantiomer. One proceeds via an *anti*- S_N2' pathway, while the other involves an *anti*- S_N2 pathway due to different interactions with the ligand during approach by the metal (Scheme 44). These results are supported by deuterium-labelling experiments, which show a relatively equal partition of S_N2 and S_N2' addition products with two different reaction rates for the formation of each and are coherent with previous studies by Pineschi and Feringa [93, 94].

This allowed the optimisation of the allylic alkylation of racemic cyclic substrates with trifluoroacetate as leaving group to provide good yields and enantioselectivities (Scheme 45).

Early 2015 Fletcher and co-workers reported the use organozirconium reagents for the dynamic kinetic asymmetric allylic alkylation of racemic 3-substituted cyclohexenes (Scheme 46) [95]. By generating in situ organozirconium reagents from terminal alkenes with Schwartz's reagent (Cp_2ZrHCl) and adding these to 3-substituted cyclohexenes, in the presence of a copper/phosphoramidite (*R,R,R*)-**L7** catalytic system, the corresponding chiral 3-alkyl-substituted cyclohexenes **118** were obtained in good yields and with excellent enantioselectivities. Interestingly, the counteranion in the copper salt proved key to the transformation with copper iodine reaching significantly higher yields than copper chloride or bromide under the same conditions. NMR studies of the potential mechanism of the reaction suggest that the iodine present in the mixture, under the influence of the copper, plays a role in the racemisation of the starting material during the reaction, potentially allowing for dynamic kinetic resolution. However, the exact role of the counteranion, as well as evidence pointing towards multiple copper ligand aggregates, makes it yet unclear as to the nature of the interactions of each enantiomer with the catalyst and whether this process is a resolution or an enantioconvergent reaction. This method provided a valuable intermediate (Scheme 46, entry **118f**) for the asymmetric total synthesis of *chaulmoogric acid* and *anthelmintic in C*.

As an extension, Fletcher and co-workers evaluated their methodology towards the performance of vinylzirconium compounds under these conditions [96]. However, despite extensive optimisation of reaction conditions, only moderate enantioselectivities could be reached (up to 19:81 e.r.). Nonetheless this demonstrated the possibility of using these sp^2 -hybridised reagents in the context of copper-catalysed allylic alkylation via dynamic kinetic resolution.

In conclusion, asymmetric allylic alkylation has evolved into one of the key catalytic enantioselective C–C bond formations. The use of hard organometallic reagents (i.e. organozinc, Grignard, organolithium) and copper catalysis, showing excellent chemo-, regio- and enantioselectivities, provides highly versatile synthetic methodology which is complementary to the Tsuji–Trost palladium-catalysed allylation with soft carbon nucleophiles. With a variety of protocols and chiral catalysts for both cyclic and acyclic substrates available, including those allowing the formation of all-carbon quaternary stereocentres, AAA has now gained a firm position in our synthetic repertoire.

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